Confidential Technical Report

Physical-Metric and Trace Certificates

for Two-Dimensional O(4) Mass Bounds

M390 in the 722-Manuscript Ram Calculus Investigation

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Abstract

We analyze the complete LaTeX source of OpenAI's preprint on sharp mass bounds for the nearest-neighbor two-dimensional O(4) model, at the catalogue-pinned public revision. Its asserted conclusions concern a full local-observable transfer gap, a unique periodic local limit, and two-sided cutoff mass bounds; they are not treated here as globally verified results. We trace the finite-volume, preliminary mixing, renormalization, calibration, matching, comparison, and continuum interfaces, including all four proof supplements.

The report's independent mathematical results are local certificates: physical-Gram positivity and residual accounting, trace-power error enclosures, a positive finite-feature approximation for the actual O(4) transfer kernel, and a compact-operator tail-resolvent criterion. We also prove a sharper square-defect recursion, with coefficient 13 on the source's threshold, in place of its relaxed coefficient 64. Exact finite tests check the algebra, an omitted-mode example, a contour certificate, and a fixed-volume perturbative coefficient. These results do not prove that the required positive-coupling square test is small and do not replace the source's nonperturbative RG estimates. Equivalent Ram coordinates neither create a spectral gap nor establish a continuum limit. All original source and private corpus files remain unchanged.

Source pin recorded by the catalogue:
`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.
Source manuscript date: September 23, 2026.
Report status: locally proved certificate development and comprehensive dependency analysis, not a certification of the entire preprint.

Contents

1 Scope, status, and the exact question 2

2 Public literature and attribution boundaries 3

3 The model, transfer kernel, and positivity 4

4 Purity controls the entire excited trace 5

5 Two-direction propagation and a local improvement 6

6 From cylinders to the full local-observable space 7

7 The preliminary mixing chain and its obligations 8

8 Linear blocking and the four RG supplements 9

9 Nonlinear exact RG: what must actually close 10

10 Calibration and the half-logarithmic scale 11

11 Admission, matching, and positive-density comparison 12

12 Assembling the source lower and upper bounds 13

13 Direct corpus correspondence and localization 14

14 Physical Gram matrices and positivity certificates 15

15 The full residual and a concrete omitted-mode test 16

16 Trace-power error bounds with the omitted space retained 17

17 A positive finite-feature approximation for the actual kernel 18

18 One-sided purity from actual transfer powers 19

19 Validated purity and square-defect intervals 20

20 A tail-resolvent count for compact transfer operators 21

21 Deterministic tests and their precise evidentiary scope 22

22 Physical units and the additional continuum hypotheses 23

23 Benefits, limitations, and a responsible next program 24

24 Read scope, provenance, and artifact contract 25

1 Scope, status, and the exact question

The object of this investigation is a particular lattice measure, not an unspecified continuum sigma model. Four internal spin components live at the sites of a two-dimensional square lattice. The source [2] asserts, for sufficiently large bare inverse coupling,

$$c\sqrt{\beta}e^{-\pi\beta} \le m_{\mathrm{lat}}(\beta) \le C\sqrt{\beta}e^{-\pi\beta}. \tag{1.1}$$

It also asserts positivity of the full gap for every finite $\beta > 0$ and uniqueness of the local limit of periodic squares. Here "full" includes rotation-invariant observables, such as bond energy. A spin-component two-point estimate does not automatically reach that sector. "Unique periodic limit" is not silently enlarged to uniqueness of every nonperiodic Gibbs state.

The local catalogue assigns applicability level 5 to manuscript 390, family 215. Its reason is the direct match between a positive integral transfer operator and the corpus's physical metric, omitted-space residual, and tail certificates. That score is an applicability assessment, not a truth probability, novelty claim, or completed proof audit. The catalogue itself explicitly withholds the small square-box inequality and the RG estimates from what the corpus establishes.

We use three distinct statuses throughout. A *source assertion* is a conclusion or estimate in the pinned manuscript whose proof chain is described here. A *local theorem* is proved in this report under its displayed hypotheses. A *checked finite instance* is an exact calculation executed by the accompanying script. No passage from the third status to a uniform infinite-volume assertion is implicit.

The public repository describes a collection at different stages of verification and warns that unformalized results may have issues [3]. We therefore call M390 a preprint, not a generally established resolution of a global open problem. Reading its complete source does not make this report a formal verification of every estimate. In particular, no Lean proof for M390 was checked, and the nonlinear RG construction was not independently implemented.

The assigned scope was the source build directory, the M390 rows of both catalogue files, and four identified private corpus files. The two volume 21 chapters were read completely; the foundation and core-patent reports were read in the relevant ranges recorded in Section 24. Only public titles, authors, identifiers, and URLs were used in web research. No private corpus text was uploaded or searched on the web.

The substantive conclusion is deliberately asymmetric. The corpus is well suited to making a finite spectral calculation honest. It does not presently supply the difficult model-specific estimate that makes the square defect small at the asymptotically free scale. The local additions below improve certification and one numerical constant in an implication, not the status of the antecedent of that implication.

2 Public literature and attribution boundaries

The public literature was revisited on October 7, 2026. Aru, Garban, and Sepulveda's paper [4], including its public introduction, describes all-temperature exponential decay for planar O(N), $N \geq 3$, as a major conjectural prediction. It proves geometric and local fluctuation results, not the full M390 conclusion. Its arXiv version is dated 2022; the source bibliography records the subsequent 2025 journal publication. This historical statement is not used to assert that no later work could exist.

Dybalski, Stottmeister, and Tanimoto [5] study the O(4), or SU(2), block-constrained variational problem in a small-field regime. Their public text specifies a field condition and estimates uniform in box size. This provides a precise constructive predecessor for local backgrounds, not an automatic proof of M390's iterated complete-density class, rough-field bounds, or terminal-coupling comparison. Their related Green-function paper [6] treats exponential localization by analytic and resolvent methods. These are ingredients at an interface; one must still prove that the particular kernels used by M390 satisfy the needed uniform bounds.

The classical reflection-positive route is attributed to Osterwalder and Schrader and to Frohlich, Israel, Lieb, and Simon [7, 8]. The source gives a discrete version of the Hilbert-space construction and uses chessboard estimates. Its finite-volume purity argument is distinct from merely quoting an infrared bound. It keeps the full trace and thereby retains multiplicities and all sectors.

The perturbative kinetic coefficients belong to the sigma-model renormalization tradition of Polyakov and Brezin, Zinn-Justin, and Le Guillou [9]. Exact continuum mass calculations by Hasenfratz, Maggiore, and Niedermayer [10] have a different logical input from the nearest-neighbor lattice problem. Volin's public paper [11] explicitly uses the integrable-theory framework and discusses its perturbative consequences. Neither that framework nor its mass formula is imported as a proof of lattice convergence or of the full periodic transfer gap.

The source also cites a companion OpenAI preprint on exponential spin decay. M390 itself correctly distinguishes its free-boundary spin-sector claim from periodic-state identification and a full local-observable gap. That companion was not independently audited here. An assertion from the same release is not independent validation of another assertion.

The browsed primary pages support the limited characterizations above. DOI links for two historical articles did not render through the browser; their bibliographic metadata were read in M390's bibliography and are identified as such in the references. The public repository was accessible, but its commit-specific tree URL failed in the browser. The exact pin in this report is therefore that of the catalogue-pinned local public snapshot; the local source snapshot has no Git metadata. Hashes bind the actual local files read, without pretending that a successful Git checkout comparison was performed.

3 The model, transfer kernel, and positivity

For even $n, w \geq 4$, let $\Lambda_{n,w} = (\mathbb{Z}/n\mathbb{Z}) \times (\mathbb{Z}/w\mathbb{Z})$. At each site $\sigma_x \in S^3$ has normalized Haar measure. The density is

$$Z_\beta(n,w)^{-1} \exp\left(\beta \sum_{\langle xy \rangle} \sigma_x \cdot \sigma_y\right) \prod_x \mathrm{d}\omega(\sigma_x), \tag{3.1}$$

with each unoriented nearest-neighbor bond counted once. Put $\mathcal{H}_w = L^2((S^3)^w, \omega^{\otimes w})$ and $V_w(s) = \sum_{i=1}^w s_i \cdot s_{i+1}$. The transfer kernel is

$$K_w(s,t) = \exp\left\{\frac{\beta}{2} V_w(s) + \beta \sum_i s_i \cdot t_i + \frac{\beta}{2} V_w(t)\right\}. \tag{3.2}$$

This is source `eq:transfer-kernel`. Integrating a cyclic product of n kernels gives $Z_\beta(n,w) = \operatorname{Tr} K_w^n$ exactly: the half-row energies combine once at each row and every time bond appears once.

Proposition 3.1 (Positive trace-class realization)**.** *For $\beta \geq 0$, K_w is a positive trace-class self-adjoint compact operator. For finite β, its top eigenvalue is positive and simple.*

Proof. Write $S = (s_1, \ldots, s_w) \in \mathbb{R}^{4w}$ and $D(s) = e^{\beta V_w(s)/2}$. Expand the crossing kernel as

$$K_w(s,t) = \sum_{r=0}^{\infty} \frac{\beta^r}{r!} \langle D(s) S^{\otimes r}, D(t) T^{\otimes r} \rangle.$$

Each term is a positive finite-rank kernel. The sum of its diagonal integrals is at most $e^{2\beta w}$, since $|S|^2 = w$ and $V_w \leq w$. Thus the series converges in trace norm to a positive operator. Uniform kernel convergence identifies it with (3.2). Symmetry is immediate. The kernel is continuous and strictly positive on a compact space. A top eigenfunction can be chosen nonnegative by replacing it with its modulus in the Rayleigh quotient. Strict positivity makes it strictly positive almost everywhere. Equality in the modulus inequality forces any top eigenfunction to have constant phase; two independent top eigenfunctions would yield an orthogonal one, impossible against the strictly positive eigenfunction. Existence follows from compact self-adjoint spectral theory and a positive Rayleigh quotient. □

Order the eigenvalues $\lambda_1 > \lambda_2 \geq \lambda_3 \geq \cdots \geq 0$. Positivity as an operator is stronger than entrywise positivity of a kernel. Both occur here, but they serve different purposes: nonnegative eigenvalues support purity and squaring; strict pointwise positivity supports simplicity of the vacuum. Replacing the physical Haar measure by arbitrary signed quadrature weights can destroy these properties.

The diagonal trace convention, with normalized product Haar measure μ_w, is explicitly

$$\operatorname{Tr} K_w = e^{\beta w} \int e^{\beta V_w(s)} \, \mathrm{d}\mu_w(s). \tag{3.3}$$

This operator trace is well-defined independently of any one-row torus convention. The physical torus partition identities above use even periods at least four. A numerical diagonal sum without its integration weights is not this trace.

4 Purity controls the entire excited trace

Suppress β and define

$$p(n,w) = 2\log Z(n,w) - \log Z(2n,w), \tag{4.1}$$
$$q(n,w) = 2\log Z(n,w) - \log Z(n,2w). \tag{4.2}$$

Axis interchange identifies $q(n,w) = p(w,n)$ and makes both nonnegative. For a fixed circumference let $a_i = (\lambda_i/\lambda_1)^n$ and $s_n = \sum_{i\geq 2} a_i$. The normalized weights are $t_i = a_i/(1+s_n)$, with $a_1 = 1$. Then

$$e^{-p(n,w)} = \sum_i t_i^2 = \frac{1+s_{2n}}{(1+s_n)^2}. \tag{4.3}$$

The trace includes every eigenvalue with its multiplicity. Nothing in this expression projects onto the vector representation of O(4).

Lemma 4.1 (Exact tail and squaring bounds)**.** *For $p = p(n,w) \geq 0$,*

$$s_n \leq e^p - 1, \qquad p(2n,w) \leq F(p), \quad F(x) = 2\log\bigl(1 + (e^x-1)^2\bigr). \tag{4.4}$$

If $p \leq 1/4$, $(\lambda_2/\lambda_1)^n \leq 2p$ and $p(2n,w) \leq 4p^2$.

Proof. Since $t_i \leq t_1$, $\sum_i t_i^2 \leq t_1 \sum_i t_i = t_1$. Thus $s_n \leq e^p - 1$. Nonnegativity gives $s_{2n} = \sum_{i\geq 2} a_i^2 \leq s_n^2$. Consequently

$$p(2n,w) = 2\log(1+s_{2n}) - \log(1+s_{4n}) \leq 2\log(1+s_n^2) \leq F(p).$$

For $p \leq 1/4$, $e^p - 1 \leq pe^p \leq 2p$ and $F(p) \leq 2p^2 e^{2p} \leq 4p^2$. The first excited term is bounded by the entire excited sum. □

This rederives source `lem:squaring`; the sharper function F is retained here for a local constant improvement. It also explains why a small individual Ritz eigenvalue error does not replace the purity test. Many small eigenvalues may have a material combined contribution.

For example, a spectrum $1, \epsilon, \ldots, \epsilon$ with M copies of ϵ has $s_n = M\epsilon^n$. A fixed bound on $\lambda_2/\lambda_1 = \epsilon$ leaves s_n arbitrarily large as M grows. Conversely, knowing only a small finite trace after omitting those states cannot bound the true purity. A trace-tail certificate is therefore the natural additional datum, not an optional diagnostic.

At $\beta = 0$ the kernel is the constant one kernel, of rank one, $Z = 1$, and $\Delta = 0$. This is an exact limiting sanity check. It says nothing about a useful low-temperature threshold at positive coupling.

5 Two-direction propagation and a local improvement

The exact rectangular identity is

$$p(n,2w) = 2p(n,w) - 2q(n,w) + q(2n,w). \tag{5.1}$$

It follows by substitution of the three partition functions on each side, without statistical independence of subrectangles. Put

$$\Delta(n) = 4\log Z(n,n) - \log Z(2n,2n) = 2p(n,n) + q(2n,n) \geq 0. \tag{5.2}$$

Area contributions cancel. Symmetry also gives $q(2n,n) = p(n,2n)$, so the test simultaneously controls a square and the already enlarged rectangle.

Theorem 5.1 (Sharper recursion on the same threshold)**.** *For the exact transfer partition functions above, if* $0 \leq \Delta(r) \leq \eta = 2^{-10}$*, then*

$$\Delta(2r) \leq 13\Delta(r)^2. \tag{5.3}$$

This improves only the coefficient in the finite-volume implication.

Proof. Set $b = p(r,2r) = q(2r,r) \leq \Delta = \Delta(r)$ and $c = p(2r,2r) \leq F(b) \leq F(\Delta)$. Interchanging directions in (5.1) gives

$$q(4r,r) = 2q(2r,r) - 2p(2r,r) + p(2r,2r) \leq 2\Delta + F(\Delta).$$

Squaring in that direction and using monotonicity of F yields

$$\Delta(2r) \leq 2F(\Delta) + F(2\Delta + F(\Delta)). \tag{5.4}$$

For $0 \leq x \leq 3\eta$, the inequalities $\log(1+y) \leq y$, $e^x - 1 \leq xe^x$, and $e^{2x} \leq (1-2x)^{-1}$ imply

$$F(x) \leq Cx^2, \qquad C = \frac{2}{1-6\eta} = \frac{1024}{509}.$$

Here $2\eta + C\eta^2 \leq 3\eta$. Substitution in (5.4) therefore bounds its right side by $[2C + C(2+C\eta)^2]\Delta^2$. Exact rational arithmetic, or direct multiplication, gives

$$2C + C(2+C\eta)^2 = \frac{1593879552}{131872229} < 13.$$

This proves the claim, and $13\eta < 1$ preserves the threshold. □

The source proves an intermediate coefficient 44 and relaxes it to 64. The improvement above uses only its finite positive-trace framework. It does not modify a single RG estimate and does not show that $\Delta_\beta(n) \leq \eta$ for any particular $\beta > 0$. If an independently certified initial defect $0 < \delta \leq \eta$ is supplied, induction gives $\Delta(2^k n) \leq 13^{-1}(13\delta)^{2^k}$. Constants in the resulting conditional gap bound can thus be tightened without changing the model-specific bottleneck.

6 From cylinders to the full local-observable space

Let $n_k = 2^k n$ and assume the initial square test with defect $0 < \delta \le \eta$. Purity gives

$$\left(\frac{\lambda_2(n_k)}{\lambda_1(n_k)}\right)^{n_k} \le 2p(n_k, n_k) \le \Delta(n_k) \le 13^{-1}(13\delta)^{2^k}.$$

Consequently all these circumferences have a common ratio bound

$$-\log \frac{\lambda_2(w)}{\lambda_1(w)} \ge \frac{c_\delta}{n} + \frac{\log 13}{w}, \qquad w = 2^k n, \tag{6.1}$$

and in particular

$$\lambda_2(n_k)/\lambda_1(n_k) \le e^{-c_\delta/n}, \qquad c_\delta = \log \frac{1}{13\delta} > 0. \tag{6.2}$$

Equation (6.1) follows by taking the negative logarithm of the preceding power bound and dividing by w. The independent audit [1] noted the corresponding refinement with 64; our proved recursion substitutes 13. These are dyadic widths only, not an estimate asserted for every even circumference. The source's weaker $c = \log 2$ also follows. If the defect is zero, purity forces rank one at the relevant widths; that case can be handled directly without writing the logarithm of zero.

For a bounded slab observable F spanning r transfer edges, its insertion kernel satisfies $|K_{w,F} f| \le \|F\|_\infty K_w^r |f|$. After division by λ_1^r, it has norm at most $\|F\|_\infty$. Separating the top eigenprojection between two slab insertions thus gives an infinite-time cylinder covariance bound

$$|\operatorname{Cov}_{\omega_w}(F, G)| \le \|F\|_\infty \|G\|_\infty e^{-c_\delta d/n}. \tag{6.3}$$

The separation d counts actual transfer edges between supports, not between arbitrarily selected block centers.

The source also identifies cylinder and square-torus limits; this is an essential step, not a consequence of fixed-width positivity alone. For a fixed insertion and $w = n_k$ sufficiently large, the difference of expectations is bounded by $2\|F\|_\infty s_{w/2}(w)$. The earlier-scale rectangle obeys $p(w/2, w) \le \Delta(w/2)$, so $s_{w/2}(w) \le e^{\Delta(w/2)} - 1 \to 0$. Products of fixed local observables have the same conclusion. Thus (6.3) passes to subsequential square limits.

In the reflection-positive Hilbert space, a centered local vector u has a positive spectral measure μ_u for the positive contraction T. If $\langle u, T^d u\rangle \le C_u a^d$ for all large d and $a < 1$, then $\mu_u((a, 1]) = 0$: any positive mass above $a + \epsilon$ would exceed the bound exponentially. Local vectors are dense after quotienting the reflection null space. Bounded spectral projections extend the exclusion to the entire vacuum complement. This establishes a full gap for these limits, including invariant sectors. Periodic-limit uniqueness remains a separate source output when identifying a single state for all sequences.

A family $T_w = \operatorname{diag}(1, e^{-1/w})$ illustrates the obstruction avoided here. Every fixed width has a positive gap, but no positive uniform lower bound survives $w \to \infty$. This example is not an O(4) counterexample; it disproves the proposed inference from fixed-width positivity alone.

7 The preliminary mixing chain and its obligations

Source `sections/preliminary.tex` is 1,284 lines and was read in full. Its theorem claims uniform all-observable mixing on unpinned tori and free subgraphs with weakened ferromagnetic bonds, with length

$$\Xi(\beta) = \exp\{\pi\beta + C\sqrt{(1+\beta)\log(2+\beta)}\}.$$

The source's derivation has the following genuinely separate stages.

First, the quaternion current identity contains both a contact term and a bracket term. For $q_y U_e q_x^{-1} = t_e + s_e$ and a left rotation, $D_\eta s_e = t_e(\eta_y - \eta_x) + (\eta_y + \eta_x) \times s_e$. The factor two in its slowly varying bracket part is quantitatively important. The pinned-cell stiffness is a contact minus a covariance, and is bounded above by the coupling; no positive lower stiffness is assumed. Compact rotation scores, buffered patches, and Whitney sums produce moment bounds uniform over the stated signed/twisted pinned class.

Second, conditioning on cell rings and augmenting with Gaussian noise gives $H_Q(A, A) = v|A|^2 - \operatorname{Var} J(A)$. Sampling estimates, conditional integration by parts, a weighted two-color estimate, and the crossed derivative estimate lead to a logarithmic variance gain. The reciprocal-square frequency sum has coefficient $1/(2\pi)$; the bracket factor converts it to the stated $1/\pi$ stiffness gain. The distinction between scalar-functional norm bounds and summed Frobenius bounds prevents an unjustified count of all modes.

Third, long and short scale steps lower the stiffness upper bound to B^{-D}. Local rotation likelihood control is upgraded from individual profiles to finite-dimensional compact parameter families. Coarse orientation variables and a disagreement exploration then give charged observable decay. The source gives a marginal-preservation argument for the adaptive exploration; this is a substantive probability argument, not a statement supplied by a coordinate transform.

Fourth, neutral observables are reached by two conditional decompositions: O(4) into two XY fields conditional on amplitudes, and XY into two Ising sign systems conditional on angles. Association and two-point control are used together with localization of *conditional means*. The identity

$$\operatorname{Cov}(F, G) = \mathbb{E}\operatorname{Cov}(F, G \mid \mathcal{A}) + \operatorname{Cov}(\mathbb{E}[F \mid \mathcal{A}], \mathbb{E}[G \mid \mathcal{A}])$$

shows exactly why discarding the second term would be an error. The source keeps it at both stages and cuts conditional bonds while retaining the original angle marginal.

Finally, uniformity under weakened bonds permits cutting a large square out of different tori. That compares local expectations, proves periodic limit uniqueness, and bounds the difference of bond energies. Integrating $\partial_\beta \Delta_\beta(n) = 8n^2[e_\beta(n) - e_\beta(2n)]$ from $\Delta_0 = 0$ gives a polynomial times $e^{-cn/\Xi(\beta)}$. This chain is mapped here, not independently certified in all constants. The Ram corpus supplies none of its nonabelian variance gain, rare-event coupling, or two-stage association estimates.

8 Linear blocking and the four RG supplements

The linear block map in source `sections/linear.tex` uses centered, smooth nonnegative block weights. For a Gaussian precision K_h, the next inverse precision is

$$K_{h+1}(p)^{-1} = a^{-1} + L^{-2} \sum_{l} |b(k_l)|^2 K_h(k_l)^{-1}, \qquad k_l = (p + 2\pi l)/L. \tag{8.1}$$

Here $a = 1$ is observation precision, not physical lattice spacing. The conditional mean P_h and covariance C_h satisfy a Schur energy identity. Its exact finite-dimensional analogue is independently checked by completing a square; the uniform analytic kernel estimates are an additional obligation.

The source requires P_h to preserve constants and centered affine fields, with fine differences gaining L^{-r} and with constants chosen before large L. Its positive edge completion introduces a curl correction which does not change $d^* B_h d$. This separation between longitudinal energy and transverse completion is central to retaining positive costs near rough fields.

The supplement `analytic-rg.tex` proves shell estimates, a common complex strip, and history differences of order L^{-h}. It also gives two analytic divisions for the edge intertwiner and an interpolation construction avoiding an extra factor L. Real-axis ellipticity alone would not imply the stated common complex strip; alias gluing and nonvanishing denominators are part of the proof obligation.

The supplement `canonical-norm.tex` fixes one exponential path norm on both sides of a block step. It keeps tagged quadratic packets $M_{r,s} - (r \cdot s)S_o$ and cancels their affine pieces before absolute values. Cubic and higher degrees gain L^{2-n}, and compensated quadratics gain L^{-1}. Its symmetry and affine hypotheses are explicit: arbitrary anchor-dependent arrays do not acquire square symmetry merely by blocking. This is a locally understandable contraction mechanism but does not prove nonlinear closure by itself.

The supplement `geometry-rg.tex` assigns each energy row and test once to a component. Its complete read sets include absence-of-seed queries, mask statuses, windows, tags, and long paths. Disjoint complete supports give exactly independent Gaussian marginals for the deliberately localized covariance. A Gaussian marginal is not the inverse of a principal precision restriction. Protectors keep logarithms in their injectivity charts, including derivative supports.

The supplement `joint-majorants.tex` treats overlapping factors under one Gaussian law. A diagonal endpoint-load matrix and a determinant bound control a product of ratio factors, followed by Holder exponents $4, 4, 2$ and explicit exceptional reserves. It does not assume independence of overlapping factors. The source verifies eight factor classes and retains single marked discrepancies even when the original factor vanishes. These are indispensable inputs to the connected expansion; individual expectation estimates cannot replace a joint product estimate.

9 Nonlinear exact RG: what must actually close

The exact observation in source `renormalization.tex` is a normalized probability kernel on S^3. It uses the normalized block mean when that mean is large enough, and a sampled constituent spin otherwise. The fallback maintains a unit direction and strict positivity; it is not discarded as a set of inconvenient configurations.

The retained density has a bulk scalar, a clipped positive kinetic part, seven canonical slots, regular errors, and a mandatory signed covering sum. The canonical slots are

$$bP_4,\ I_2,\ bP_5,\ I_3,\ bP_6,\ I_4,\ b^{-1}J_2. \tag{9.1}$$

Quadratics have their affine Hessians removed into the running coupling. Odd invariant tensors are included where permitted; an unjustified even-parity truncation would change the canonical class. The error norm has eight field derivatives; discrepancies use seven so that changing a map does not accumulate a derivative loss at every scale.

The source chooses reference layers $H_j = H + \gamma j$, a mesoscopic scale $M_j = \lceil (\log H_j)^2 \rceil$, logarithmic p_j, and thresholds $t_j = H_j^{-1/2} p_j$, $T_j = H_j^{-0.49}$. Geometry uses these fixed reference thresholds, not a moving exact coupling. This matters for continuity under shooting and for comparing two histories on common label lists.

The energy ledger splits the exact kinetic difference into fine rows, observation rows, output corrections, mean mismatch, nonnegative reserve, and truncation defect. Rough components pay a positive reserve. Gaussian completion on their complement retains determinant and inverse-residual chains explicitly. Smooth charts, attached protectors, and transported hard inputs delimit where differentiation is legitimate.

Prepared-factor products are summed into connected components with complete supports. Only the output-empty components enter the Mayer logarithm; output-seeded components remain in a mandatory covering gas. Mask strengthening is compensated by an exact covering correction. These operations are identities of the density, not informal deletion of rare terms. A signed gas is allowed, but the actual full endpoint density is strictly positive.

Closure asserts both renewed norm caps and an extracted coupling

$$b' = b + \alpha_h(\boldsymbol{P}) + \kappa_h(\boldsymbol{P})/b + \Delta_h, \qquad |\Delta_h| \le C_L H_j^{-1.05}. \tag{9.2}$$

Shape differences contract forward, while coupling differences include a small term proportional to the coupling discrepancy. Bulk constants and kinetic coefficients are period-independent; winding errors remain in the retained lists. The finite order of choosing constants is essential: uniform free estimates first, then L, caps and weights, then logarithmic powers and H.

This report does not replace the 1,496-line closure proof with a claim that "positive matrices contract." Positivity of a finite Gram matrix addresses a different object. The source's rough-field reserves, uniform support sums, Taylor remainders, and parameter comparisons still require their own proofs at every admitted scale.

10 Calibration and the half-logarithmic scale

Source `calibration.tex` determines canonical kicks by evaluating the same finite torus partition function directly and after blocking. Its direct Laplace expansion is taken at fixed volume. Uniformity in depth enters only through previously proved canonical coefficient and history bounds. Exchanging these two roles would be invalid.

For spins on S^D, with $v = n^2$, mean-zero lattice Green function G_x, $G = G_0$, and $r = (1-v^{-1})/2$, its first coefficient is

$$A(n) = \frac{Dvr^2}{4} - \frac{D(D-1)}{2}G. \tag{10.1}$$

The source derives it from the tangent Gaussian action and the mean-axis Jacobian. The companion script independently constructs the rational Green matrix at $n = 4$, $D = 3$, and verifies this coefficient from the displayed Gaussian moment formula. This is a meaningful normalization check, but it verifies neither the two-loop asymptotics nor RG closure.

At the next order the source decomposes the first interaction into second and fourth Gaussian chaoses. Discrete product rules express the variance through j_0, j_1 and Green-function terms. A logarithmic finite-size correction to j_0 survives doubling after cancellation of volume and quadratic Green terms. Comparing with the blocked expansion then identifies

$$\alpha_h = -\gamma + O(\rho^h), \qquad \kappa_h = -\frac{\gamma}{2\pi} + O(\rho^h), \qquad \gamma = \frac{\log L}{\pi}.$$

Their interpretation depends on the original convention that every bond is counted once and β is inverse coupling.

Proposition 10.1 (Conditional summation of the drift)**.** *Suppose an admitted sequence with $b_0 = \beta$, ending in a fixed compact interval, obeys*

$$b_{h+1} = b_h - \gamma - \nu/b_h + r_h + e_h, \quad |r_h| \le C b_h^{-1-\epsilon}, \quad \sum_h |e_h| \le C,$$

with $\nu = \gamma/(2\pi)$, sufficiently large b_h, and bounded visits to each unit coupling interval, where $\epsilon > 0$ and all bounds are uniform in the starting value β. Then $N \log L = \pi\beta - \frac{1}{2}\log\beta + O(1)$.

Proof. Set $\Phi(b) = b/\gamma - (\nu/\gamma^2)\log b$. Taylor expansion over a bounded step cancels the two b^{-1} terms and gives $\Phi(b_{h+1}) - \Phi(b_h) = -1 + O(b_h^{-1-\epsilon'}) + O(|e_h|)$, where $\epsilon' = \min(1, \epsilon)$. Bounded visits make the power errors summable. Telescoping and using the fixed endpoint proves the result. □

In the source the bounded-visit fact follows by summing a negative drift with a globally summable exceptional term. The proposition explains the scale algebra without claiming those bounds anew. Losing the half logarithm would lose the prefactor $\sqrt{\beta}$, not merely a harmless bounded constant.

11 Admission, matching, and positive-density comparison

In source `trajectories.tex`, j counts remaining steps and the free history is $N - j$. The nearest-neighbor input begins with zero canonical and regular terms; rough energy differences populate its covering gas. Admission is established by a first-exit argument around $H_j + (2\pi)^{-1}\log(H_j/H)$. Exact terminal values are obtained by a continuous shooting argument. It asserts existence, not uniqueness, of the selected bare coupling $\beta_N(h)$.

Comparing the final K steps of depths $N \geq K$ gives a mixed boundary problem: shape data have bounded discrepancy at the initial end, while the kinetic couplings agree at the terminal end. With a contracting shape inequality and sufficiently small coupling feedback, geometric weighted maxima bound both discrepancies by $C(1+K)^C \rho^{K-j}$. This is not comparison at equal bare coupling; arbitrarily separated bare couplings are allowed precisely because the terminal kinetic values are matched.

Source `comparison.tex` addresses a separate obstacle. Its endpoint representations contain a signed covering sum Ξ. Absolute coefficient closeness does not bound Ξ_2/Ξ_1 when the denominator is small. The proof uses one-law chessboard rarity, editable rough regions on S^3, complete-density finite energy, exact removal and inclusion–exclusion identities, and a forest sum on a common good set. Positivity is assigned only to complete sums, never to restricted signed backgrounds. Both laws must give the common exceptional set small probability.

Lemma 11.1 (Normalization needs both laws)**.** *Let $f_i > 0$, $Z_i = \int f_i$, $\mu_i = f_i/Z_i$, and suppose $\mu_i(G^c) \leq \epsilon_i < 1$ and $|\log(f_2/f_1)| \leq a$ on G. Then*

$$e^{-a}(1-\epsilon_1) \leq Z_2/Z_1 \leq e^a/(1-\epsilon_2).$$

Proof. The ratio of integrals over G is in $[e^{-a}, e^a]$, and $Z_2/Z_1 = (\int_G f_2/\int_G f_1)\mu_1(G)/\mu_2(G)$. Use $1-\epsilon_i \leq \mu_i(G) \leq 1$. □

This elementary lemma is source `cmp:normalization`, rederived for clarity. Control under only one law is insufficient because the other may concentrate outside G. The full source comparison gives a bound of order $C_H e^{-d_H K}$ only within a prescribed volume window $M^2 \leq e^{c_H K}$. The restriction is part of the theorem, not a technicality that may be dropped in an infinite-volume limit. The corpus's finite spectral bounds do not prove the rarity or signed-sum comparison hypotheses.

12 Assembling the source lower and upper bounds

After exact blocking, the source writes

$$Z_{\beta_r(h)}(ML^r, ML^r) = e^{v_r(h)M^2}\mathcal{Z}_{r,h}(M).$$

Since $v_r(h)$ is independent of the coarse period, its contribution cancels from $D_r(h;M) = \Delta_{\beta_r(h)}(ML^r)$. Comparing at both M and $2M$ yields

$$|D_N(h;M) - D_K(h;M)| \le C_H e^{-d_H K}, \qquad (2M)^2 \le e^{c_H K}. \tag{12.1}$$

Both boxes must lie in the allowed window. Matching terminal coupling and removing bulk scalars are separate requirements.

The preliminary length exceeds L^K by a subexponential factor: $\log(\Xi(\beta_K)/L^K) = O(\sqrt{K\log K})$. Choose dyadic M_K with $\log M_K = K^{3/4} + O(1)$. Then the preliminary decay overwhelms its polynomial prefactor while $2\log(2M_K) = o(K)$ remains within the comparison window. One finite K_0 can be selected, and then $M_0 = M_{K_0}$ is frozen. Only after this choice does N grow.

This ordering yields the source lower bound

$$m_{\mathrm{lat}}(\beta_N(h)) \ge \frac{\log 2}{M_0 L^N}.$$

Using Theorem 5.1 instead changes its conditional finite-box constant, not the existence of a useful explicit K_0 or M_0. No numerical values of these RG-dependent constants are certified here. Their mere formal finiteness is not an executable low-temperature test.

The upper bound has a different dependency. One-law rarity gives $\mathbb{E}|V_X - V_{X+e}|^2 \le \epsilon_H$. Along three bonds, $\mathbb{E}|V_0 - V_{3e}|^2 \le 9\epsilon_H$. For $\epsilon_H \le 1/9$, O(4) invariance gives $\mathbb{E}[V_0^{(1)} V_{3e}^{(1)}] \ge 1/8$. Pull back by

$$F_X(\sigma) = \mathbb{E}[V_X^{(1)} \mid \sigma].$$

Distinct ancestral blocks use conditionally independent auxiliary variables, so their pulled-back correlation equals the terminal one. This does not assert independence of F_X and F_Y under the original spin law. Each is centered, bounded by one, and supported in its own ancestral block of side L^N.

Across an actual support separation at least L^N, the full transfer gap bounds the correlation by $e^{-m_{\mathrm{lat}}L^N}$. Thus $m_{\mathrm{lat}} \le (\log 8)L^{-N}$. This argument needs a nonzero surviving observable and the norm bound in the full reflected space. It does not follow from convergence of individual finite-cylinder eigenvalues. Calibration then changes the depth scale to $\sqrt{\beta}e^{-\pi\beta}$, conditional on the complete source chain.

13 Direct corpus correspondence and localization

The foundation report [12] proves exact equivalence of a global integral transform and its displacement-indexed Ram form under absolute integrability. Its inverse localization is a coordinate inverse for the kernel, not an inverse for the integral operator on states. That distinction survives intact in this application.

The row space $(S^3)^w$ is not a Euclidean interval. A subtraction $s - t$ in ambient $\mathbb{R}^{4w}$ does not furnish global intrinsic coordinates. Choose a finite atlas $\phi_a : U_a \to \mathbb{R}^{3w}$ and a smooth partition of unity χ_a subordinate to it. Input and output charts are independent: set $x = \phi_a(s)$ and $y = \phi_b(t)$, with input Haar density $j_b(y)$. The product-chart contribution is

$$(K_{ab}f)(\phi_a^{-1}(x)) = \chi_a(\phi_a^{-1}(x)) \int_{\phi_b(U_b)} h_{ab}(y, x-y) f(\phi_b^{-1}(y))\, \mathrm{d}y \tag{13.1}$$

$$= \chi_a(\phi_a^{-1}(x)) \int_{x-\phi_b(U_b)} h_{ab}(x-\alpha, \alpha) f(\phi_b^{-1}(x-\alpha))\, \mathrm{d}\alpha, \tag{13.2}$$

where

$$h_{ab}(y, \alpha) = K(\phi_a^{-1}(y+\alpha), \phi_b^{-1}(y))\chi_b(\phi_b^{-1}(y))j_b(y).$$

The domain is $y \in \phi_b(U_b)$ and $y + \alpha \in \phi_a(U_a)$. Outside U_a define the output contribution as zero. The identity $K = \sum_{a,b} K_{ab}$ follows from $\sum_a \chi_a(s) = \sum_b \chi_b(t) = 1$, so arbitrary cross-chart pairs are included. The difference $\alpha = x - y \in \mathbb{R}^{3w}$ is a coordinate device, not an intrinsic displacement or a translation symmetry of the spin manifold. An individual nonsymmetric chart contribution need not be positive; positivity belongs to the reconstructed whole unless a symmetric factorization is supplied.

Compactness and boundedness ensure integrability here. A Taylor or moment truncation of the local expression adds a remainder, which must be bounded in a physical operator or trace norm. Derivatives at a point are not defined for an arbitrary L^2 state. Thus a jet ansatz requires its own regularity class and cannot be inferred from the Hilbert-space notation alone.

The physical-metric chapter [13] supplies injective synthesis J, the physical Gram matrix $G = J^*J$, and the orthogonal residual split. We specialize these directly to $\mathcal{H}_w$ below. The spectral-limits chapter [14] supplies counterexamples to discarding the complement and a domain-aware inverse criterion. Its unbounded indefinite pollution example is a warning about inference, not a claim that every compact positive compression has that same pollution mechanism.

The core report's spectral gate [15] controls a Schur complement on a contour and requires the tail to have no enclosed spectrum. Its stated compact-resolvent setting is not literally the setting of a compact transfer operator on an infinite-dimensional space. Section 20 proves the required adaptation away from zero instead of silently invoking incompatible hypotheses.

14 Physical Gram matrices and positivity certificates

Choose linearly independent trial functions $\psi_1, \dots, \psi_d$ in $\mathcal{H}_w$ and let $Jc = \sum c_i \psi_i$. Write

$$G = J^*J > 0, \quad H = J^*K_wJ, \quad B = G^{-1}H, \quad U = JG^{-1/2}, \quad P = UU^*. \tag{14.1}$$

Then U is isometric and P is the orthogonal projection onto the trial space. The symbol H here is a finite matrix, not the source's terminal coupling; the latter is always indicated by context or a subscript.

Theorem 14.1 (Metric-preserving finite representation)**.** *The matrices in* (14.1) *satisfy* $H \geq 0$, $GB = B^*G$, *and*

$$A_P = U^*K_wU = G^{-1/2}HG^{-1/2} \geq 0.$$

Moreover $G^{1/2}BG^{-1/2} = A_P$. *Consequently* $\operatorname{Tr} B^n = \operatorname{Tr} A_P^n$ *for every positive integer* n.

Proof. For c, $c^*Hc = \langle Jc, K_wJc\rangle \geq 0$. Symmetry gives $H = H^*$. Multiplying $B = G^{-1}H$ by G gives the metric symmetry. The formula for A_P follows by substituting U. It is a similarity of finite matrices, so powers have the same trace. The physical norm is $\|Jc\|^2 = c^*Gc$, not c^*c. □

The theorem concerns a compression, not the full transfer trace. It does not assert $K_wJ = JB$, which would be an invariant-space identity. One may compute a nonsymmetric-looking B accurately without having broken physical symmetry, provided the metric is retained. Conversely, simply replacing B by $(B + B^*)/2$ changes the operator in general.

Proposition 14.2 (A usable approximate-metric condition)**.** *If Hermitian* $\widehat{G}$ *and a certified scalar* $g_0 > 0$ *satisfy* $\widehat{G} \geq g_0 I$ *and* $\left\|G - \widehat{G}\right\| \leq e < g_0$, *then*

$$(1 - e/g_0)\widehat{G} \leq G \leq (1 + e/g_0)\widehat{G}.$$

Proof. $-eI \leq G - \widehat{G} \leq eI$ and $I \leq g_0^{-1}\widehat{G}$. Add $\widehat{G}$ to these inequalities. □

This turns absolute quadrature and rounding error into a relative norm certificate only after a lower bound for the physical metric has been proved. Near-dependent trial functions may make g_0 very small. Extra arithmetic precision cannot repair a rank-deficient quadrature rule: sampling at q points gives rank at most q. Basis selection, preconditioning, quadrature accuracy, and omitted-state control must therefore be tracked as separate operations.

15 The full residual and a concrete omitted-mode test

Theorem 15.1 (Exact physical residual account)**.** *For any $c \in \mathbb{C}^d$ and $\lambda \in \mathbb{C}$,*

$$\|(K_w - \lambda)Jc\|^2 = \|(B - \lambda)c\|_G^2 + \|(I - P)K_w Jc\|^2 . \tag{15.1}$$

For real λ and $Jc \neq 0$,

$$\operatorname{dist}(\lambda, \operatorname{spec} K_w) \leq \frac{\|(K_w - \lambda)Jc\|}{\|Jc\|}.$$

Proof. $PK_wJ = JB$ and $(I - P)J = 0$. The two components of the residual are $J(B - \lambda)c$ and $(I - P)K_wJc$, which are orthogonal. Pythagoras gives (15.1). The spectral theorem bounds the squared full residual below by the squared distance to the spectrum times $\|Jc\|^2$. □

This is a direct specialization of the private physical-metric result, with no new claim of discovery. It makes explicit which residual the O(4) certificate would have to calculate. A small full residual locates *some* spectral value; it does not identify it as the first excited one or prove the absence of intervening spectrum. A spectral count or global norm/tail argument is still needed.

The exact finite test takes

$$A = \operatorname{diag}(1, 1/3, 1/7), \qquad J = \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad G = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

Here $J^*AJ = \left(\begin{smallmatrix} 8/7 & 1 \\ 1 & 4/3 \end{smallmatrix}\right)$ and $B = G^{-1}J^*AJ$ is not Euclidean symmetric. For $c = (2, -1)^T$ and $\lambda = 2/5$, the script verifies all three terms in (15.1) exactly: the full squared residual is $6934/11025$, the projected part is $8702/33075$, and the omitted part is $484/1323$. The omitted term is nonzero.

For an even simpler logical obstruction, retain only the first two coordinates of a positive three-dimensional matrix with a nonzero coupling to the third. A retained exact eigenvector can have zero projected residual while its full residual is entirely the coupling into that third coordinate. No accuracy of the finite eigensolver changes this fact.

For the actual compact positive K_w, increasing dense orthogonal projections does have favorable fixed-width convergence properties. But their rates need not be uniform in w or β. The report uses this structure where valid and does not transplant the corpus's indefinite spectral-pollution example as if it were a counterexample to compact positive variational theory. The common lesson is the need to identify the omitted space and quantify it.

16 Trace-power error bounds with the omitted space retained

The finite trace needed in purity is different from a single residual. The following certificate converts a whole-operator error into the specific nonlinear observable used by M390.

Theorem 16.1 (Trace-norm propagation)**.** *Let $A, \widetilde{A}$ be positive trace-class operators on the same Hilbert space, with norms at most M and $\left\|A-\widetilde{A}\right\|_1 \le \epsilon$. Then for every integer $n \ge 1$,*

$$|\operatorname{Tr} A^n - \operatorname{Tr} \widetilde{A}^n| \le nM^{n-1}\epsilon. \tag{16.1}$$

For an orthogonal finite-rank projection P, put $A_0 = PAP$, $Q = I - P$, and suppose $\operatorname{Tr}(QAQ) \le \tau$, $\|QAP\|_1 \le \rho$. Then $\|A - A_0\|_1 \le \tau + 2\rho$ and

$$0 \le \operatorname{Tr} A^n - \operatorname{Tr} A_0^n \le n\,\|A\|^{n-1}\,(\tau + 2\rho). \tag{16.2}$$

Proof. The noncommutative telescoping identity is $A^n - \widetilde{A}^n = \sum_{j=0}^{n-1} A^{n-1-j}(A-\widetilde{A})\widetilde{A}^j$. The trace ideal inequality bounds each summand's trace norm by $M^{n-1}\epsilon$. For the compression, write the difference in $P \oplus Q$ blocks as the off-diagonal blocks plus QAQ. Positivity makes the latter trace norm equal its trace. Triangle inequality proves the bound. Ordered compression eigenvalues are at most the corresponding eigenvalues of the positive compact operator by the max-min characterization. Summing their nth powers gives the lower inequality. □

An alternative bound uses a computable Hilbert–Schmidt leakage: $\|QAP\|_1 \le \sqrt{d}\,\|QAP\|_2$, where $d = \operatorname{rank} P$. If only the positive diagonal tail trace is known, factorization $QAP = (QA^{1/2})(A^{1/2}P)$ gives

$$\|QAP\|_1 \le \sqrt{\operatorname{Tr}(QAQ)\operatorname{Tr}(PAP)}. \tag{16.3}$$

These follow from Cauchy–Schwarz for singular values and the Schatten product inequality, respectively. They expose the price of not measuring the coupling to the complement directly.

The bounds may be conservative. Scaling to a positive contraction removes exponential factors M^{n-1} but not relative error when $\operatorname{Tr} A^n$ itself is tiny. Correlated errors, sharper eigenvalue enclosures, or feature truncations can improve efficiency. No such improvement is assumed without its own certificate.

A trace-norm bound at fixed w is not uniform control of all widths. To use the source's finite criterion, certify the initial full square partition functions and invoke the exact model's rectangular identities. Do not construct unrelated truncated matrices at different widths and assume that their approximate partition functions obey those identities.

17 A positive finite-feature approximation for the actual kernel

Define a scaled kernel $A_w = e^{-2\beta w} K_w$. On its diagonal, $A_w(s,s) = e^{\beta(V_w(s)-w)} \le 1$, so $\operatorname{Tr} A_w = e^{-\beta w} \int e^{\beta V_w} \, \mathrm{d}\mu_w \le 1$ and $\|A_w\| \le 1$. Truncate its tensor series at degree m to obtain

$$A_{w,m}(s,t) = e^{-2\beta w} D(s) D(t) \sum_{r=0}^{m} \frac{\beta^r}{r!} (S \cdot T)^r. \tag{17.1}$$

Theorem 17.1 (Certified positive feature tail)**.** *For every $\beta \ge 0$ and finite w, m,*

$$0 \le A_{w,m} \le A_w, \qquad \|A_w - A_{w,m}\|_1 \le \epsilon_{w,m} := e^{-\beta w} \sum_{r>m} \frac{(\beta w)^r}{r!}. \tag{17.2}$$

Thus $0 \le \operatorname{Tr} A_w^n - \operatorname{Tr} A_{w,m}^n \le n\epsilon_{w,m}$. If $x = \beta w < m+2$, a rational-function upper bound is

$$\epsilon_{w,m} \le \frac{x^{m+1}}{(m+1)!} \frac{1}{1 - x/(m+2)}. \tag{17.3}$$

Proof. Each tensor degree is positive, so the tail is positive in operator order. Its trace norm is its diagonal integral. The squared feature length is $|S|^{2r} = w^r$, while $D(s)^2 \le e^{\beta w}$. This proves (17.2). Apply Theorem 16.1 with contraction norm one, and use ordered eigenvalues for the sign. For $r \ge m+1$, successive scalar series terms have ratio at most $x/(m+2) < 1$. Sum the geometric majorant and drop $e^{-x} \le 1$. □

In fact the diagonal calculation gives the exact refinement

$$\operatorname{Tr}(A_w - A_{w,m}) = (\operatorname{Tr} A_w)\epsilon_{w,m}.$$

The same factor $\int e^{\beta V_w} \, \mathrm{d}\mu_w$ occurs in every degree, so division by the full diagonal trace gives precisely the displayed Poisson tail. The inequality in (17.2) uses only $\operatorname{Tr} A_w \le 1$. This is a total one-step trace statement, not a uniform relative-error bound for $\operatorname{Tr} A_w^n$ at growing widths.

The finite feature space has dimension at most $\sum_{r=0}^{m} \binom{4w+r-1}{r} = \binom{4w+m}{m}$. Sphere relations create dependencies, so this is only an upper bound. The nonzero eigenvalues of the feature operator equal those of its feature Gram matrix: if Φ synthesizes weighted monomials, $A_{w,m} = \Phi\Phi^*$ and the matrix is $\Phi^*\Phi$. Null dependencies must be removed or handled on the positive quotient.

This construction preserves all sectors represented by the features and bounds the total omitted trace analytically. It is a genuine model-specific certificate opportunity. It is not a completed numerical O(4) mass computation: feature integrals still contain $e^{\beta V_w}$, and their quadrature needs rigorous error control. The rank grows combinatorially, especially at the large widths relevant to low temperature. The conservative trace-tail estimate may be unusable there without additional structure. The script evaluates exact upper bounds for three finite parameter choices, not the source's small-square condition.

18 One-sided purity from actual transfer powers

The independent bounded purity audit [1] supplied the following useful refinement. It is rederived here to make its scope and its distinction from a projected-power calculation explicit.

Theorem 18.1 (One-sided omitted-trace certificate)**.** *Let $T \geq 0$ be trace class, let J be injective with finite-dimensional domain, and put $G = J^*J$, $P = JG^{-1}J^*$, $Q = I - P$, and*

$$C = G^{-1/2}J^*TJG^{-1/2}, \quad S_P = \operatorname{Tr} C > 0, \quad R_P = \operatorname{Tr} C^2.$$

If $\operatorname{Tr}(QTQ) \leq \tau$, then

$$\frac{\operatorname{Tr} T^2}{(\operatorname{Tr} T)^2} \geq \frac{R_P}{(S_P + \tau)^2}, \qquad p(T) := 2\log \operatorname{Tr} T - \log \operatorname{Tr} T^2 \leq U(T) := 2\log(S_P + \tau) - \log R_P. \tag{18.1}$$

No invariance of the trial space is required.

Proof. The orthogonal block decomposition gives $\operatorname{Tr} T = S_P + \operatorname{Tr}(QTQ) \leq S_P + \tau$ and

$$\operatorname{Tr} T^2 = R_P + 2\,\|QTP\|_2^2 + \operatorname{Tr}((QTQ)^2) \geq R_P.$$

All terms are finite since a positive trace-class operator is Hilbert–Schmidt. Divide the lower numerator bound by the upper denominator bound and take the negative logarithm. The isometry $JG^{-1/2}$ identifies the retained block with C. □

For the actual scaled kernels take $T_1 = A_n^n$ and $T_2 = A_{2n}^n$, with independently certified retained data and tail bounds. Exact axis interchange gives

$$\Delta(n) = 2p(T_1) + p(T_2) \leq 2U(T_1) + U(T_2). \tag{18.2}$$

Thus an upper bound at most 2^{-10} certifies the original square test. With $t_i = \tau_i / S_{P,i}$ and $p_{P,i} = 2\log S_{P,i} - \log R_{P,i}$, its right side is $2p_{P,1} + p_{P,2} + 4\log(1 + t_1) + 2\log(1 + t_2)$. For rank-one retained blocks and $t_i \leq t$, the sufficient tolerance is $t \leq e^{1/6144} - 1$. This is a required error budget, not an achieved O(4) computation.

This certificate needs no off-diagonal estimate because leakage can only increase its numerator. It does need the retained matrix of the *actual power*: generally $PA_w^nP \neq (PA_wP)^n$. The latter replacement is not licensed by (18.1). Certified lower bounds for R_P and upper bounds for $S_P + \tau$, including Gram and quadrature errors, can be substituted with the same inequality direction.

There is a concrete tail connection. If P contains the feature space of $A_{w,m}$, then $QA_{w,m} = 0$. Because $0 \leq A_w \leq I$, $A_w^n \leq A_w$ by spectral calculus, and hence

$$\operatorname{Tr}(QA_w^nQ) \leq \operatorname{Tr}(QA_wQ) \leq \operatorname{Tr}(A_w - A_{w,m}) \leq \epsilon_{w,m}.$$

This supplies a valid absolute tail for the one-sided test at fixed β, w, n. Relative accuracy and the exact-power integrals may still be expensive. Neither this inequality nor the finite-feature theorem asserts a uniform tail estimate over unbounded widths.

The exact check script verifies (18.1) for the coupled rational contraction from Section 21, using actual powers 1, 2, and 4 and a nonorthogonal retained basis. It also checks that the projected square differs from the square of the projection.

19 Validated purity and square-defect intervals

Suppose rigorous arithmetic, quadrature, and omitted-space estimates produce $0 < L_n \le Z(n,w) \le U_n$ and $0 < L_{2n} \le Z(2n,w) \le U_{2n}$. Then monotonicity of the logarithm immediately gives

$$2\log L_n - \log U_{2n} \le p(n,w) \le 2\log U_n - \log L_{2n}. \tag{19.1}$$

The lower endpoint may be replaced by its maximum with zero for the true model. Its purity lies in $[L_{2n}/U_n^2, U_{2n}/L_n^2]$, intersected with $[0,1]$.

For the square test let $L_n^{\mathrm{sq}} \le Z(n,n) \le U_n^{\mathrm{sq}}$ and $L_{2n}^{\mathrm{sq}} \le Z(2n,2n) \le U_{2n}^{\mathrm{sq}}$. A sufficient certified gate is

$$4\log U_n^{\mathrm{sq}} - \log L_{2n}^{\mathrm{sq}} \le 2^{-10}. \tag{19.2}$$

This is a bound for the *actual* partition functions. If its upper endpoint exceeds the threshold, the test is unresolved unless a lower endpoint also exceeds it. Failure of this conservative test is not proof that the model has no gap.

Scaling each kernel by $e^{-2\beta w}$ changes $\log Z(n,w)$ by $-2\beta n w$. The p, q, and Δ combinations cancel this area term exactly. Therefore the contraction scaling in Section 17 can be used without changing the tests. The same cancellation holds for any consistently tracked bulk free-energy term, but not for arbitrary width-dependent renormalizations.

Proposition 19.1 (Relative error budget)**.** *If $\widehat{Z}_j > 0$ and $|Z_j/\widehat{Z}_j - 1| \le e_j < 1$, then*

$$|\log Z_j - \log \widehat{Z}_j| \le -\log(1-e_j).$$

For square sides $n, 2n$, the defect error is at most $4[-\log(1-e_n)] + [-\log(1-e_{2n})]$.

Proof. $1 - e_j \le Z_j/\widehat{Z}_j \le 1 + e_j$, and $\log(1+e_j) \le -\log(1-e_j)$. Apply triangle inequality to the linear combination defining the defect. □

This displays the actual precision target. Large individual free energies almost cancel, so a small absolute error in a visually large number can still overwhelm the 2^{-10} gate. Interval arithmetic must include metric errors, trace errors, quadrature errors, and final logarithm evaluation. An unvalidated floating-point decimal is not an interval endpoint. Exact rational tests in this report verify the underlying algebra but are not a substitute for such a positive-coupling partition-function enclosure.

Proposition 19.2 (A certified neighborhood in bare coupling)**.** *For even $n \ge 4$ and $\beta, \beta_0 \ge 0$,*

$$|\Delta_\beta(n) - \Delta_{\beta_0}(n)| \le 16n^2|\beta - \beta_0|. \tag{19.3}$$

Consequently a certificate $\Delta_{\beta_0}(n) \le d_0 < \eta$ certifies the same threshold whenever $|\beta - \beta_0| \le (\eta - d_0)/(16n^2)$ and $\beta \ge 0$.

Proof. On the n-square torus there are $2n^2$ bonds, each with interaction in $[-1,1]$. Differentiating its finite-volume logarithmic partition function gives an expected energy of absolute value at most $2n^2$. On the $2n$ torus the bound is $8n^2$. The derivative of their combination $4\log Z(n,n) - \log Z(2n,2n)$ therefore has absolute value at most $4(2n^2) + 8n^2 = 16n^2$. Integrate between the two couplings and apply the certified margin. □

This refinement is also in the independent audit. It is local in coupling and depends on the chosen box. It supplies neither an initial successful positive-coupling test nor sharp-scale RG stability. At $d_0 = \eta$ it gives no positive radius. The conditional dyadic gap bound and the distinctions about periodic limits remain unchanged.

20 A tail-resolvent count for compact transfer operators

Decompose a compact self-adjoint A using a finite orthogonal projection:

$$A = \begin{pmatrix} A_P & C^* \\ C & D \end{pmatrix} \quad \text{on } P\mathcal{H} \oplus Q\mathcal{H}.$$

The tail D is compact and bounded; on an infinite-dimensional tail it does not have compact resolvent. Zero is an unavoidable spectral accumulation point. The following formulation repairs that mismatch with the private compact-resolvent gate while preserving its proof idea.

Theorem 20.1 (Compact-operator Schur certificate)**.** *Let Γ be a simple closed contour bounding a region $\mathcal{U}$ whose closure does not contain zero. Suppose $D - z$ is invertible on a neighborhood of $\overline{\mathcal{U}}$, $A_P - z$ is invertible on Γ, and*

$$\sup_{z \in \Gamma} \left\| (A_P - z)^{-1} C^* (D - z)^{-1} C \right\| < 1. \tag{20.1}$$

Then A and A_P have the same total eigenvalue multiplicity in $\mathcal{U}$, and A has no spectrum on Γ.

Proof. For $0 \le t \le 1$, replace C by tC. Block elimination reduces invertibility to $A_P - z - t^2 C^*(D - z)^{-1} C$, invertible by a uniform Neumann series. All interpolated operators are compact. Their spectral projections inside a contour bounded away from zero have finite rank. The resolvent identity and the uniform inverse bound make those Riesz projections norm-continuous in t. Projections within norm distance less than one have equal rank, so the rank is constant. At $t = 0$ the tail has no enclosed spectrum and only the finite block contributes. □

For example take $A_P = 4/5$, $D = \operatorname{diag}(1/(5j))_{j \ge 1}$, and couple the retained coordinate only to the first tail coordinate with strength $1/100$. On $|z - 4/5| = 1/10$, the distance to the entire tail spectrum is at least $1/2$. Thus the gate is at most $10 \cdot 2 \cdot 10^{-4} = 1/500 < 1$. The coupled operator has exactly one eigenvalue in the disk. The example is positive and compact; positivity of the coupled two-dimensional block follows from its positive diagonal and determinant. All remaining tail entries are positive.

This certifies a count in a separated region, not a gap above the vacuum unless the remaining interval is also excluded. A grid of contour samples is insufficient without a uniform between-node bound. Resolvent derivatives or certified Lipschitz estimates can provide that bound. Likewise, tail invertibility only on the contour does not exclude uncounted tail eigenvalues inside. The full-region premise is intentional.

21 Deterministic tests and their precise evidentiary scope

The accompanying `check_m390.py` uses exact rational SymPy and Python fraction arithmetic. There is no randomized sampling and no private network access. Its results are written to `_audit/M390-author/checks.json`. Assertions are executable checks, not a separate proof of arbitrary untested cases.

Check group	What it establishes
Physical metric	Exact $GB = B^*G$, positive Gram determinant, and the orthogonal residual identity for a rational positive operator.
Compression trace	Exact trace powers for a coupled positive contraction, and verification against the trace-tail plus off-diagonal budget.
One-sided purity	Actual-power retained traces in a nonorthogonal basis; the omitted-trace bound and failure of projected-power substitution.
Purity	Rational spectral weights for three finite spectra and powers 2, 4, 8; the total tail and its squaring inequality.
Recurrence constant	Exact rational verification of the constant in Theorem 5.1 and its domain/threshold inequalities.
O(4) feature tail	Three exact scalar geometric majorants for the actual kernel feature tail; no partition function evaluated.
Schur example	The full infinite diagonal tail separation and the exact 1/500 gate for the declared compact operator.
Finite calibration	The $n = 4$, $D = 3$ first Laplace coefficient from an exact 16-site Green matrix and Gaussian moments.

The compression test uses

$$A = \begin{pmatrix} 3/4 & 0 & 1/20 \\ 0 & 1/4 & 0 \\ 1/20 & 0 & 1/8 \end{pmatrix}, \qquad P = \operatorname{diag}(1, 1, 0).$$

Principal minors certify both $A > 0$ and $I - A > 0$. The trace-tail budget is $1/8 + 2/20 = 9/40$, valid but deliberately conservative. Powers $1, 2, 4, 8, 16$ are evaluated exactly. The test exercises a nonzero omitted coupling, rather than an invariant diagonal truncation where the residual problem would disappear.

For finite calibration the Green matrix is $G = (L + \Pi_0)^{-1} - \Pi_0$, where L is the 16-site torus Laplacian and Π_0 projects onto constants. The script compares the direct expectation of the displayed first interaction with (10.1). It does not estimate the large-n logarithmic correction or the second coefficient by fitting a curve. The exact results are $G_0 = 103/384$, $G_{e_1} = 13/384$, and $A(4) = 469/256$.

All tests are reproducible with `py -3 check_m390.py`. Rendering uses the separately installed, existing MiKTeX and Poppler through the read-only shared export helper. Mathematical test output, build logs, page contacts, and final hashes have different evidentiary roles and are retained separately. A successful TeX run says nothing about the truth of the model-specific RG premises.

22 Physical units and the additional continuum hypotheses

Fix a positive reference length ℓ_{ref} independently of the mass and set $a_N = \ell_{\mathrm{ref}} L^{-N}$. Conditional on the source's depth bounds, the physical cutoff mass satisfies

$$\frac{\log 2}{M_0 \ell_{\mathrm{ref}}} \le \frac{m_{\mathrm{lat}}(\beta_N)}{a_N} \le \frac{\log 8}{\ell_{\mathrm{ref}}}. \tag{22.1}$$

This is not a choice of units made by setting the unknown mass to one. The unit is fixed by block scale and terminal coupling. Neither convergence of the sequence of masses nor existence of a continuum measure follows from its two-sided boundedness.

Proposition 22.1 (Conditional continuum lower gap)**.** *Suppose positive lattice contractions T_j have centered gaps m_j, $a_j \to 0$, and $\liminf m_j/a_j \ge m_* > 0$. Suppose a limiting positive self-adjoint generator H with vacuum Ω has a dense set of centered vectors u with lattice approximants u_j satisfying*

$$\|u_j\|^2 \to \|u\|^2, \quad a_j n_j(t) \to t, \quad \langle u_j, T_j^{n_j(t)} u_j\rangle \to \langle u, e^{-t\mathsf{H}} u\rangle$$

for every $t \ge 0$. Then the vacuum complement has spectrum in $[m_*, \infty)$ and contains no zero-energy vector.

Proof. The lattice quadratic form is at most $e^{-(m_j/a_j)a_j n_j(t)} \|u_j\|^2$. Passing to the limit gives $\langle u, e^{-t\mathsf{H}} u\rangle \le e^{-m_* t} \|u\|^2$. Density extends this inequality to the complement. Positive spectral measures then exclude energies below m_*, including zero. □

This is the source's conditional continuum proposition rederived here. Its alternative positive-time regularization replaces a possibly divergent bare vector norm by convergence of a positive-time autocorrelation. It still requires a dense family and limiting semigroup matrix elements. A coordinate identity provides none of those convergence statements.

An upper bound additionally needs observable survival. If centered continuum vectors f, g of norms at most one satisfy $|\langle f, e^{-d\mathsf{H}} g\rangle| \ge 1/8$ at $d \ge \ell_{\mathrm{ref}}$, the continuum gap is at most $(\log 8)/\ell_{\mathrm{ref}}$. Cutoff low-energy states can lose all weight in a chosen limit: $j^{-1}\delta_1 + (1 - j^{-1})\delta_R \to \delta_R$ for $R > 1$. Thus bounded cutoff spectral bottoms alone do not prove that a limiting observable sees them. Euclidean covariance and the interpretation as a relativistic mass remain further requirements.

23 Benefits, limitations, and a responsible next program

The useful contribution of Ram coordinates here is an accountable representation of the same physical operator. Nonorthogonal functions can be adapted to local kernels or symmetries, with their physical Gram matrix retained. Full residuals expose leakage. Positive feature tails and trace bounds turn a retained computation into an explicit interval for the quantity the finite-volume theorem actually uses. The recurrence improvement is a proved local gain, with an exact constant check and unchanged small-defect regime.

The costs are substantial. Trial dimensions grow rapidly with width, Gram matrices may be ill-conditioned, and absolute trace estimates can be weak in the small relative-error regime. The actual transfer kernel is an operator on a continuous spin space; no arbitrary finite sampling is automatically a faithful Haar discretization. Sector-specific calculations cannot be promoted to a full gap without a complement argument. Uniformity in width and inverse coupling must not be inferred from success at a fixed pair of parameters.

The source's global assertions remain dependent on its preliminary nonabelian mixing, exact nonlinear closure, two-loop calibration, admitted trajectories, and positive-density comparison. All were read; not all were independently reconstructed. In particular the present tests do not establish $\Delta_\beta(n) \le 2^{-10}$ for any positive coupling at a sharp-scale box size. No claim that the corpus resolves the all-temperature lattice conjecture is made.

A practical next program should start with a modest fixed circumference and a symmetry-aware feature Gram matrix, rigorous integration, and positive tail bounds. It should record interval endpoints for the actual $Z(n, n)$ and $Z(2n, 2n)$ and test (19.2). A successful finite certificate can then enter the exact rectangle theorem. An unsuccessful one should diagnose whether the limitation is feature rank, integration error, or genuine lack of a small test at that size.

For the RG chain, independent audits should target statements with distinct obligations: the adaptive orientation coupling's marginals; both conditional-mean localization errors; same-norm canonical contraction and its affine cancellation; hard-factor transport on derivative supports; complete read-set factorization; finite-period scalar extraction; and the forest encoding for signed relative comparison. Rechecking these interfaces is more informative than repeating the final scale algebra.

The report deliberately does not rank a large unverified theorem as proved merely because its endpoint is expressible in equivalent coordinates. It gives smaller fully derived tools and states exactly what an additional model-specific proof or validated computation would have to provide. That is both the main benefit and the present limit of the source/corpus correspondence.

24 Read scope, provenance, and artifact contract

The source root is the local directory ending in `Sharp-mass-bounds-for-the-two-dimensional-O4-model-September-23-2026/build`. All its files were read: `main.tex` (80 lines), `references.bib` (382), the four figure sources (159 lines total), and the following substantive files. Counts describe the read snapshot.

Section file	Lines	Supplement file	Lines
introduction	552	analytic-rg	316
finite-volume	213	canonical-norm	278
preliminary	1284	geometry-rg	466
linear	319	joint-majorants	294
renormalization	1496		
calibration	405		
trajectories	290		
comparison	928		
assembly	151		
upper	142		
physical	114		

The manuscript README and repository README were also read. The M390 rows of `applicability.json` and `manuscripts.json` were parsed as structured data. No assertion of reading all 722 manuscripts is made.

The directly read private scope is: volume 21 `ch01d-physical-metrics.tex`, lines 1–210, complete; volume 21 `ch01f-spectral-limits.tex`, lines 1–242, complete; foundation `TR-rev-1-old-tr117.tex`, lines 191–366; and core `TR-core-pat-1.tex`, lines 855–1045. The last range includes the spectral theorem, contour control, worked example, and adjacent release conditions. No unread theorem elsewhere in those larger private reports is represented as checked.

The separately authored bounded audit [1] was read in full before the final formatting round. It found no error in the finite-volume argument under its stated conventions, while explicitly excluding full preliminary and RG verification. Its actual-power one-sided certificate is attributed and proved in Section 18; its dyadic-width and periodic-state scope agrees with the distinctions used here. The audit's own script was not rerun by this author. Our separate exact checks cover the locally incorporated formula. Its report hash is recorded as a read input, not as an author-owned output.

The report and script are new files in `M390-O4-Mass-Bounds`. Author-owned audit outputs are under `_audit/M390-author`, with the named interim findings file beside it. The shared export helper writes only the named `_build/TR-M390-Ram-O4-Mass/round-1` and `round-2` trees for this report. It was read but not edited. No catalogue, queue, progress file, source original, or other author's artifact is changed by this work.

The native editor open request was queued, with visibility unconfirmed; native compilation failed with "Unable to find standard directories for platform." Both were attempted with bounded waits, and the exact outcomes are recorded in the audit metadata. The separately requested on-disk PDF is exported with installed MiKTeX and Poppler, without installation. At most two formatting rounds are used, with all-page contacts and detailed page views. The final SHA256 manifest binds the delivered source, PDF, check script, checks, read-scope record, and audit outputs. The manifest's own digest is not recursively included. Source hashes are evidence of the local read snapshot, not proof of global validity of its mathematics.

References

[1] Independent bounded M390 purity/transfer audit. *M390: Independent bounded purity / transfer audit*, October 7, 2026. Confidential local artifact `_audit/M390-independent-purity-audit.txt`, read completely. Finite-volume, normalization, and conditional continuum checks; full nonlinear RG and preliminary mixing expressly outside its scope. Section 8 supplies the one-sided actual-power certificate incorporated here with its own proof and local exact tests.

[2] OpenAI. *Sharp mass bounds for the two-dimensional O(4) model*. September 23, 2026, preprint. Catalogue-pinned local public snapshot at revision `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Pinned public manuscript. Complete local LaTeX build read; no global verification claimed.

[3] OpenAI. *Math release repository and README*. https://github.com/openai/math. Public page accessed October 7, 2026; local README read completely. Repository verification disclaimer applies to the collection; manuscript-specific formal verification was not checked.

[4] J. Aru, C. Garban, A. Sepulveda. *Percolation for 2D classical Heisenberg model and exit sets of vector valued GFF*. Communications in Mathematical Physics 406 (2025), article 37. https://arxiv.org/abs/2212.06767; https://arxiv.org/html/2212.06767v1. Public abstract and introduction inspected October 7, 2026. Journal metadata from the source bibliography.

[5] W. Dybalski, A. Stottmeister, Y. Tanimoto. *The Balaban variational problem in the nonlinear sigma model*. Reviews in Mathematical Physics (2024), 2461003. https://arxiv.org/abs/2403.09800; https://arxiv.org/html/2403.09800v1. Public abstract, introduction and small-field contraction passages inspected October 7, 2026.

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[11] D. Volin. *From the mass gap in O(N) to the non-Borel-summability in O(3) and O(4) sigma-models*. https://arxiv.org/abs/0904.2744. Public abstract inspected October 7, 2026. Published as Physical Review D 81 (2010), 105008, with the journal title recorded in the source.

[12] M. Subbarao. Private foundation corpus, *TR-rev-1-old-tr117*, local LaTeX report. `RamFoundationTRs/TR-rev-1-old-tr117/TR-rev-1-old-tr117.tex`. Read lines 191–366, especially `eq:RT`, `eq:RLT`, `eq:IRLT`, and the proposition on equivalence of global and Ram forms. Confidential, not independently web-published by this report.

[13] M. Subbarao. Private *RamCalBooks*, volume 21, chapter file `ch01d-physical-metrics.tex`, complete 210-line read. Locators: `v21c01:physical-gram-definition`, `v21c01:metric-symmetry`, `v21c01:residual-decomposition`, `v21c01:gram-equivalence`. Local snapshot hashed in read-scope record.

[14] M. Subbarao. Private *RamCalBooks*, volume 21, chapter file `ch01f-spectral-limits.tex`, complete 242-line read. Locators: `v21c01:pollution-proposition`, `v21c01:strong-not-norm`, and "Domain-aware approximate inversion." Local snapshot hashed in read-scope record.

[15] M. Subbarao. Private core report `RamPatsCore50-TRs/TR-core-pat-1/TR-core-pat-1.tex`. Read lines 855–1045, including `sec:spectral`, `thm:spectral`, contour sampling and the worked tail-resolvent example. The compact transfer adaptation is proved separately in this report.